

PROJECT PROPOSAL

YummyNaKha!

A Personalized Menu App for Reading, Translating, and Safely Choosing Thai Dishes

“Don’t just translate the menu. Find what’s yummy for you.”

Item	Detail
Team	5-Idiots
Project title	YummyNaKha! — Personalized Menu App
Document	Project Proposal (DISCOVER phase deliverable)
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1. Executive Summary

YummyNaKha! is a personalized menu application for people who cannot easily read Thai restaurant menus. A diner saves a food profile once — favorite foods, plus allergies and doctor-advised food restrictions under a separate opt-in — then uploads a photo of a Thai-language menu. The system extracts each dish name and price, translates the menu from Thai into English, infers the likely ingredients of each dish, checks every dish against the diner's profile, and presents the menu split into three categories: recommended, check before ordering, and conflict. The diner selects dishes and opens a separate page that shows those dishes back in Thai, so they can order by showing the screen to restaurant staff.

Basic translation is not enough on its own. It shows what a dish is called, but not whether it matches a diner's taste, nor whether it is likely to conflict with an allergy or a doctor-advised restriction. Working that out dish by dish — with a translation app, guesswork, or by asking staff across a language barrier — is slow and uncertain, so diners fall back on a few familiar dishes, avoid unfamiliar restaurants, or order without resolving the uncertainty.

This proposal describes a single, tightly scoped workflow — profile in → menu photo → extract and translate → match against the profile → three categories → order in Thai — delivered over a 14-week programme by a five-member team. The system is designed under Thai law from the first sprint: allergy and doctor-advised dietary data is treated as sensitive personal data under the PDPA and collected only under an explicit, separate opt-in; every content action is logged under the Computer Crime Act §26; and every acceptance and consent is stored as a retrievable, versioned record under the Electronic Transactions Act.

How this proposal reads evidence. Three kinds of statement are kept apart on purpose: (a) validated user evidence from the survey of six respondents (Sep 6–7, 2026), used in §2; (b) team-defined engineering targets, marked as such throughout §3, which are proposed numbers for review, not user-research results; and (c) items marked TBD, which are not yet known and are not filled in by assumption.

Success will be measured, not asserted. The project metric is the time (and number of steps) a diner needs to find and select suitable dishes from a Thai menu, compared between the diner's current method and YummyNaKha!, measured with 15+ real users in the Test phase. No baseline exists yet, so no percentage improvement is claimed in this proposal — the target is set once a baseline is measured (NFR1).

2. Problem Statement

Diners who cannot read Thai struggle to choose suitable food from a Thai-only menu. These pains were confirmed in a survey of six respondents (U1–U6, Sep 6–7 2026) and are referenced throughout this proposal as P1–P5.

ID	Pain	Evidence
P1	Difficulty understanding unfamiliar Thai menu information	All six survey respondents report encountering menus they do not fully understand — one almost every time, one often, four sometimes. Five of six requested translation and four of six requested prices, and one respondent reports sometimes ordering anyway without resolving the gap.
P2	Dependence on other people and tools to decide	Reported methods are asking restaurant staff, asking a friend, looking at pictures of the dish, and Google Translate / Google Search — or falling back on a dish already known.
P3	Wanting help matching dishes to preferences and restrictions	Expressed as requests for recommendations based on favorite foods, saved favorite foods and possible allergy warnings, and highlighting detected allergen concerns. The survey did not record whether these respondents personally have an allergy or a doctor-advised restriction — the request is for the capability, not a self-reported medical status.

ID	Pain	Evidence
P4	Accuracy and trust decide whether a diner will rely on the app	“Accuracy” and “Is it really trustworthy and accurate?” are the two most direct concerns raised; a third respondent would hesitate to rely on the app if information is inaccurate, there are not enough reviews, or the app is difficult to use.
P5	Wanting a direct, photo-based way to process a menu	Three of six respondents requested taking or uploading a menu photo; one requested nothing else.

The consequence is a weekly loss of both time and confidence: diners default to a few familiar dishes, avoid unfamiliar restaurants, or order without resolving an allergy or restriction concern — the exact behaviour a doctor-advised restriction is meant to prevent.

3. Project Goals and Objectives

Goal. Help a diner understand a Thai menu and choose a suitable dish quickly — personalized to their taste, and honest about what the app cannot verify from a photo.

3.1 Objectives

- **Make the menu readable.** Extract and translate every dish and price from an uploaded Thai menu photo into English.
- **Choose faster.** Reduce the time and number of steps a diner needs to find and select suitable dishes, compared with their current method.
- **Personalize the result.** Rank and flag dishes using the diner’s saved favorite foods, allergies, and doctor-advised restrictions.
- **Warn, never certify.** Flag dishes that may conflict with an allergy or restriction as items to check with restaurant staff — never as a guarantee that a dish is safe.
- **Make ordering easy.** Show the diner’s selected dishes back in Thai on a separate page for ordering.
- **Comply by design.** Satisfy PDPA, Computer Crime Act §26, and Electronic Transactions Act obligations as build requirements, not as a later audit.

3.2 Success criteria

ID	Criterion	Target
NFR1	Project metric — decision time and steps	TBD — measured with 15+ real users; no baseline exists yet, so no percentage improvement is claimed until one is measured
NFR2	Extraction accuracy	≥ 90% dish-name extraction; ≥ 95% price extraction where a clearly visible price is present [engineering target]
NFR3	Translation quality	≥ 90% of extracted dish names / descriptions judged acceptably translated [engineering target]
NFR4	Matching fidelity — the safety-critical figure	Missed known conflicts ≤ 5% (reported on its own); false conflict warnings ≤ 10% [engineering target]
NFR5	End-to-end latency	≤ 10 seconds for 90% of supported scans, upload to categorized result [engineering target]
NFR6	Usage / cost ceiling	TBD — provider fixed (Google Gemini API); cap pending the cost model and quota decision

ID	Criterion	Target
NFR7	Availability	≥ 95% during scheduled user-testing and demo periods [MVP operational target, not a commercial SLA]; failures shown in plain language, never a raw error
NFR8	Ease of use	TBD — target set from the first usability test
NFR9	Traffic-log retention	≥ 90 days where the CCA §26 obligation applies, stored separately from content

4. Scope

4.1 In scope — the core workflow

The system delivers one workflow end to end: profile in → menu photo → extract + translate → match → three categories → order in Thai. Ten Must-have areas and one Should-have support it.

#	Feature	Description	Priority
1	Food Profile	Save favorite foods and ordinary preferences.	Core
2	Allergy & Restriction Opt-in	Record allergies and doctor-advised restrictions under a separate, explicit opt-in.	Core
3	Thai Menu Image Upload	Upload a photo of a Thai-language menu; no live camera scanning.	Core
4	Dish & Price Extraction	Pull each dish name and its price from the image.	Core
5	Thai → English Translation	Translate the extracted menu text.	Core
6	Ingredient Inference	Infer likely ingredients per dish, to enable matching.	Core
7	Personalized Recommendations	Check every dish against the saved profile.	Core
8	Three-Category Result	Label every dish recommended / check before ordering / conflict.	Core
9	Dish Selection & Order in Thai	Select dishes; view them back in Thai on a separate page.	Core
10	Consent, Disclosure & Data Rights	Terms, AI disclaimer, separate transfer consent, withdrawal, account deletion.	Core
11	Multiple Menu Images	Upload more than one menu image in a session.	Should

4.2 Out of scope for this phase

- Scan history — reopening a previously scanned menu. Menu images are never kept as permanent history (Won't — team decision, LR5).
- Live camera scanning of a menu — upload only (Won't).
- Voice output of dishes or results (Won't).
- Any language pair other than Thai → English (Won't).
- Restaurant-side registration, setup, or supplied menu data (Won't).
- Rating a recommendation, so the team can improve accuracy over time (Could — not core).
- Any “certified”, “nutritionist-approved”, “doctor-approved”, or “100% accurate” claim — excluded by LR11.
- Any statement that a dish is safe, and any safety score or percentage — excluded by LR12.

- Any permanent storage of menu images — excluded by LR5.

5. Feature Breakdown and Acceptance Criteria

Requirements say what a diner wants; features say what gets built. Each feature below is testable — a QA tester can pass or fail every criterion. The full acceptance-criteria list (FE1–FE9, with individual AC1–AC8 rows) lives in the backlog and requirement spec; this table condenses each feature to its key, testable commitments.

ID	Feature	Key acceptance criteria	Traces
FE1	Account & Food Profile	Create an account and sign in with OAuth or email/password, with the profile present on the next session. Ordinary preferences and the sensitive allergy/restriction section are handled separately, with the sensitive section behind an explicit opt-in — never a silent default. Any profile field, including the sensitive section, can be viewed, edited, and deleted. Email/password accounts store a salted hash only; OAuth stores a token only.	F1–F4 · LR1, LR6
FE2	Consent & Disclosure Gate	Terms, privacy notice, and the AI & Accuracy Disclaimer must be accepted before the first menu is processed. Consent to send data to the third-party provider — Google Gemini API — is a separate action, states what is processed, and is required before any categorized result. Every acceptance and every withdrawal is stored, retrievable by user ID.	F15, F16 · LR2, LR9
FE3	Menu Capture & Extraction	Upload a photo of a Thai-language menu (upload only, no live camera). Dish-name extraction $\geq 90\%$ and price extraction $\geq 95\%$ on the defined test set. The outbound request to Gemini carries menu text and needed profile flags only — no account email, name, or identifier — and every call is made from the backend. Multiple images per session are supported (Should).	F5, F6, F19 · LR3
FE4	Menu Translation	Extracted Thai dish names and descriptions are translated into English; $\geq 90\%$ judged acceptably translated on the defined evaluation set. Thai $\rightarrow$ English is the only supported direction.	F7
FE5	Ingredient Inference & Profile Matching	Likely ingredients are inferred for each dish and checked against the saved profile, labelled recommended / check before ordering / conflict. Incomplete or uncertain menu information always resolves to check — never recommended. Missed known conflicts $\leq 5\%$, false conflicts $\leq 10\%$ on the controlled test set; the pipeline completes within the latency target for 90% of scans.	F8–F10 · LR12
FE6	Result Presentation & Safety Wording	Every check and conflict result tells the diner to confirm with restaurant staff. No result ever states a dish is safe — the only permitted clear-result phrasing is that no conflicting ingredient was detected from the available menu information. No safety score or percentage is ever shown. Every result shows its basis: the extracted text, its translation, and the profile item behind a check or conflict.	F11, F12 · LR11, LR12
FE7	Selection & Order in Thai	Select dishes from the categorized menu; a separate page lists the selection with dish names in Thai, legible enough for restaurant staff to read from the screen, reachable without re-processing the menu. No voice output.	F13, F14
FE8	Failure & Degradation Handling	An unreadable photo, a non-Thai-menu photo, or a provider outage/rate-limit produces a plain-language message, never a raw error; after a failed extraction the uploaded image stays on screen so the diner can review, crop, and retry without re-photographing the menu. The app targets $\geq 95\%$ availability during scheduled testing and demo periods.	F18, F26 · LR5

ID	Feature	Key acceptance criteria	Traces
FE9	Data Rights & Compliance Layer (cross-cutting)	Account deletion removes the food profile, any temporary image, and results. The uploaded image is retained only for the active scan session and is never kept as permanent history. No data flow exists to an advertising or profiling sink. Traffic data is retained ≥ 90 days where CCA §26 applies, stored separately from content — profiles and images are not over-retained merely because of that rule. Given a scan ID, the stored AI-processing record reproduces what the diner was shown.	F17 · LR4, LR5, LR7, LR8, LR10, LR11

Build order for the core workflow: FE1 → FE2 → FE3 → FE4 → FE5 → FE6 → FE7, with FE8 and FE9 running behind all of them.

6. Proposed Solution and Architecture

6.1 System architecture

The application is a mobile-first web client over a service-oriented backend and four separate data stores. Compliance sits in the call graph rather than in policy documents alone — the architecture below is the team's proposed design for BUILD, translating the legal and functional requirements of §9 into concrete components.

Layer / component	Responsibility
Client (mobile-first web app)	Screens S1–S10: sign-up, taste profile, allergy opt-in, upload, consent, processing, results, select, order-in-Thai, profile & data.
Profile Service	Food-profile CRUD; separates ordinary preferences from the sensitive allergy/restriction section (F1–F4).
Menu Processing Service	Upload handling, dish/price extraction, translation, ingredient inference (F5–F8).
Matching Service	Checks each dish against the profile and applies the uncertainty-resolves-to-check rule (F9–F12).
Consent Service	Terms/disclaimer acceptance, the separate transfer consent, and withdrawal (F15, F16).
AI Gateway	The single outbound chokepoint to the third-party AI service: strips identity, enforces the usage cap, and is the only place holding the provider key (LR3, LR6d, NFR6).
Audit & Log Service	Writes account ID, IP, and timestamp for every create/edit/delete action (LR4, LR7).
App DB	Food profile and dietary data, encrypted at rest (PDPA — sensitive data).
Consent store	Append-only acceptance and consent records, including withdrawals (LR9).
Access / traffic log store	Account ID, IP, and timestamp, retained ≥ 90 days, kept separate from content (LR7).
AI-processing record store	Model/version, OCR result, translation, profile snapshot, and timestamp — an internal quality measure (LR10).

6.2 Key design decisions

- **The AI Gateway is its own component.** One chokepoint where identity is stripped, the usage cap is enforced, and the provider key lives — compliance in one testable place instead of scattered through the app (LR3, LR6d, NFR6).

- **Four stores, not one.** Access logs must survive content deletion (LR7, LR8) and consent must be append-only (LR9); different lifecycles cannot share a table.
- **The Consent Service gates the AI Gateway.** An ungated path from the app to the third-party AI service does not exist in the call graph (LR2).
- **The dietary profile is encrypted at rest.** Allergies and doctor-advised restrictions are sensitive personal data under the PDPA (LR1).
- **The uploaded menu image is temporary, never permanent.** It lives for the active scan session only, so it can be reviewed, cropped, or retried, then it is deleted — there is no image store that outlives a session (LR5, F26).

6.3 Stated trade-off: third-party AI service instead of a self-hosted model

We gain: a working extraction, translation, and matching pipeline early in Build with no model-training or GPU-hosting work, at output quality good enough to test the real hypothesis — that personalized, translated, allergy-aware menu reading reduces decision time and dependence on others. A five-person team with a fixed course schedule cannot train and ship a self-hosted model and still deliver the workflow.

We pay: per-call cost and rate limits (NFR6 cap still TBD), an availability dependency the team does not control (NFR7 sets a 95% MVP target and requires a plain-language failure message, never a raw error), and a PDPA cross-border transfer obligation that requires disclosure, consent, and data minimisation before every first send (LR2, LR3).

Revisit when: generation cost per active diner exceeds a sustainable ceiling once the Gemini cost model and quota are settled (NFR6), or a missed-conflict failure traces to model behaviour the team cannot constrain by prompt (the NFR4 safety-critical figure).

7. User Journey

The diner in this journey — “D” — is not a named persona or a single respondent but an anonymised composite assembled only from traits the survey actually recorded: eats Thai food several times a week or more, meets a menu they do not fully understand sometimes to almost every time, currently asks staff or a friend or uses a translation app, and will not rely on a result unless it seems accurate and trustworthy. Nationality, residency, student status, age, and whether D personally has an allergy are not assumed — the survey captured none of these.

#	Screen	D does	System does	Traces
J1	Sign-up / Terms	Creates an account (OAuth or email/password); accepts Terms, privacy notice, AI & Accuracy Disclaimer.	Stores each acceptance: user ID, document version, timestamp, action.	— (trust foundation)
J2	Taste profile	Saves favorite foods and ordinary preferences.	Stores as normal personal data — not behind the sensitive opt-in.	P3
J3	Allergy opt-in	Chooses whether to add allergies / doctor-advised restrictions, in a separate section.	Nothing is set until explicit opt-in; editable and deletable afterwards.	P3
J4	Menu upload	Uploads a photo of a Thai-language menu.	Accepts the image; upload only, no live camera.	P5, P2

#	Screen	D does	System does	Traces
J5	Transfer consent (first scan only)	Reads what is sent and consents separately from the Terms already accepted.	Discloses the Google Gemini API receives menu text and needed profile flags only — never account email or real name; the call is made from the backend.	P4
J6	Processing	Waits, with the uploaded photo still on screen.	Extracts dish/price → translates → infers ingredients → checks against the profile; shows progress within the latency target.	P1
J7	Categorized result	Reads the menu in English, split three ways.	Every dish labelled recommended / check / conflict; uncertain items are always check, never recommended; every check/conflict carries the confirm-with-staff instruction; each result shows its basis.	P1, P3, P4
J8	Dish selection	Picks the dishes to order.	Collects the selection.	P1, P5
J9	Order-in-Thai page	Opens the selected-menu page and shows the screen to staff.	Lists chosen dishes back in Thai, legibly, without re-processing the menu.	P2
J10	Ongoing control	Later: edits the profile, withdraws consent, or deletes the account.	Applies the edit; stores a withdrawal the same retrievable way; deletion removes profile, temporary image, and results (traffic logs excepted where CCA §26 applies).	— (data rights)

7.1 Alternate and failure paths

- The photo is unreadable (blur, glare, angle): a plain-language message says what to do next; the image stays on screen for crop or retry.
- The image is not a Thai-language menu: reported as such rather than processed into meaningless output.
- The provider is down or rate-limited: a plain-language notice, never a raw error.
- No dish qualifies as recommended: the check and conflict lists still render; no dish is promoted to fill an empty category.
- The diner declines the transfer consent: no data is sent, no result is produced, and the reason is explained.
- The diner withdraws consent after using the app: processing stops; the withdrawal is stored the same retrievable way as the consent.
- The diner never opts in to allergies: matching runs on ordinary preferences only — no allergy claim is made in either direction.
- The menu runs across several pages: the diner uploads more than one image in the session (Should).
- A dish is extracted with no visible price: the price is left empty, never guessed.
- The diner deletes the account: profile, temporary image, and results are removed; traffic logs are retained separately where CCA §26 applies.
- The diner abandons the scan or closes the app mid-session: the temporary image is deleted when the session ends; nothing is kept as history.

8. Prototype

A low-fidelity wireframe set fixes what appears on each screen and why; colour, type, and spacing are the Designer's decision during BUILD. No production code is written in the DISCOVER phase.

Screen	Name	Purpose	Traces
S1	Sign up & Terms	Create a persistent identity; capture the three acceptances (Terms, privacy notice, AI & Accuracy Disclaimer).	F1, F15 · J1
S2	Taste profile	Ordinary, non-sensitive preferences — favorite foods, spice level, texture.	F2 · J2
S3	Allergy & medical opt-in	The sensitive section: a single off-by-default toggle, explicit opt-in, nothing pre-set.	F3, F4 · J3
S4	Home / Upload	The one primary action — a large upload target filling most of the screen.	F5 · J4
S5	Transfer consent	Shown once before the first generation: exactly what is sent and what is not.	F15 · J5
S6	Processing	Photo panel pinned at top; stepped progress; recovers into a plain-language failure state.	F6–F9, F26 · J6
S7	Results	The whole menu, in English, split three ways, with a summary strip and filter chips.	F9–F12 · J7
S8	Select / review	Compact list of only the selected dishes, removable, with running total.	F13 · J8
S9	Order in Thai	Minimal, high-contrast, one dish per row, built for a staff member reading at arm's length.	F14 · J9
S10	Profile & data	Ongoing control: edit preferences, edit the allergy section, view/withdraw consents, delete account.	F4, F16, F17 · J10

System prototype (Figma): [Execute action — YummyNaKha! prototype](#), built by the Designer (Yoon Nadi). Every screen in the file is labelled with its S# and the F / LR ids above so the prototype traces back to this spec.

<-----Paste exported prototype screenshots or high-fidelity mockups from the Figma file here----->

9. Legal and Compliance Requirements

Three Thai laws govern this system. Each obligation below is carried into the backlog as a Must item, so compliance is verified by acceptance testing rather than by review at the end.

9.1 Personal Data Protection Act (PDPA)

- Allergies, medical conditions, and doctor-advised restrictions are treated as sensitive personal data, collected only under explicit, separate opt-in consent — never as a silent default (LR1).
- Sending pantry-equivalent menu items and dietary flags to a third-party AI service is disclosed and consented to before the first generation, and minimised to ingredient/menu text plus dietary flags — never account email, name, or household identifiers (LR2).
- Every outbound request to the Google Gemini API carries only the menu text and the profile flags the recommendation needs; every call is made from the backend, never the frontend (LR3).
- Profile and menu data are used only for menu understanding and personalized recommendation — never advertising, resale, or profiling — unless separately opted in (LR4).
- An uploaded menu image is kept only for the active scan session, never as permanent history; account deletion removes the profile, temporary image, and results within the stated window (LR5).
- Email/password accounts store a salted password hash only, never plaintext; OAuth stores a token only; the Gemini API key is held server-side only (LR6).

9.2 Computer Crime Act §26

- Creating content (a manual entry, a saved result, a shared plan) logs the account ID, IP address, and timestamp (LR7).
- Edit and delete actions log actor identity, IP, and timestamp, kept ≥ 90 days even after the item itself is deleted (LR7).
- Food profiles, the temporary scan-session image, and selected dishes are not retained 90 days merely because of §26 — their retention follows the PDPA schedule instead (LR8).
- Log retention on cloud infrastructure is configured explicitly to ≥ 90 days, since cloud defaults are usually shorter (NFR9).

9.3 Electronic Transactions Act §9 / 26 / 28

- Accepting the Terms and the AI & Accuracy Disclaimer records user ID, timestamp, and document version as a retrievable acceptance record (LR9).
- Consent for sensitive dietary data and for the AI transfer — and its withdrawal — is stored as a retrievable, timestamped, versioned record (LR9).
- Every AI-generated result stores its exact text, the model version, the pantry/menu snapshot, and the dietary filters used, so the output can be reproduced in a later dispute (LR10).
- The UI never uses “certified”, “nutritionist-approved”, “doctor-approved”, or “100% accurate” language, because no licensed certifying party or professional is involved (LR11).

9.4 Product safety rule

- No screen, message, or result ever states that a dish is safe for a diner’s allergy or restriction. The only permitted clear-result phrasing is that no conflicting ingredient was detected from the available menu information. No safety score or percentage is ever shown — the three categories only. Every check and conflict result tells the diner to confirm with restaurant staff (LR12).

10. Project Team and Responsibilities

Team 5-Idiots is a five-member team. Each member owns one area end to end and is accountable for the deliverables listed against that role.

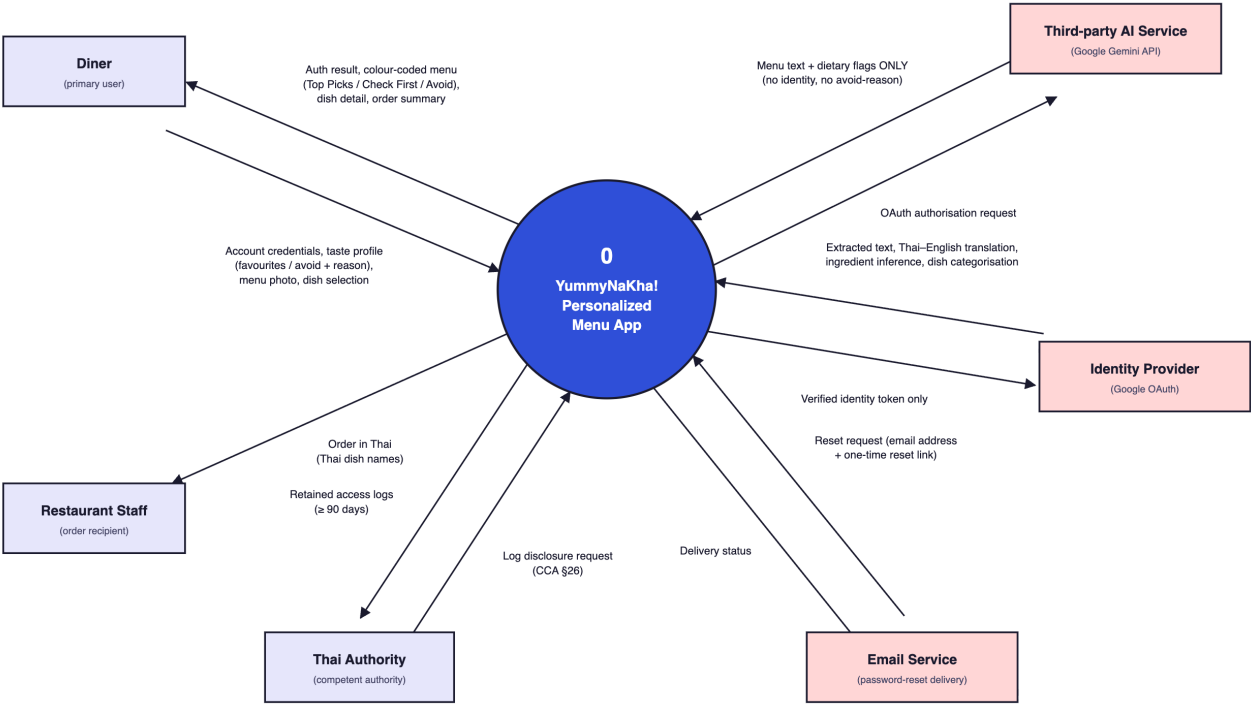
Student ID	Name	Role	Responsibilities
6631503089	Tay Zar Tun	Product Owner	Owens the project vision; defines requirements; manages scope, priorities, and timeline.
6631503052	Aike Paung Bra	QA / Tester	Verifies specs against the backlog; checks traceability and legal requirements; tests that acceptance criteria are met.
6631503055	Chan Nyein Thu	AI Lead	Owens the team's AI tooling; builds the agents/skills, the spec, and the backlog; handles AI disclosure and governance.
6631503093	Thiri Kyaw Khaing	Tech Lead	Oversees the GitHub repository, code standards, and technical decisions.
6631503099	Yoon Nadi	Designer	Creates user-friendly, visually appealing, intuitive digital experiences; owns the wireframes and the visual prototype.

10.1 Other stakeholders

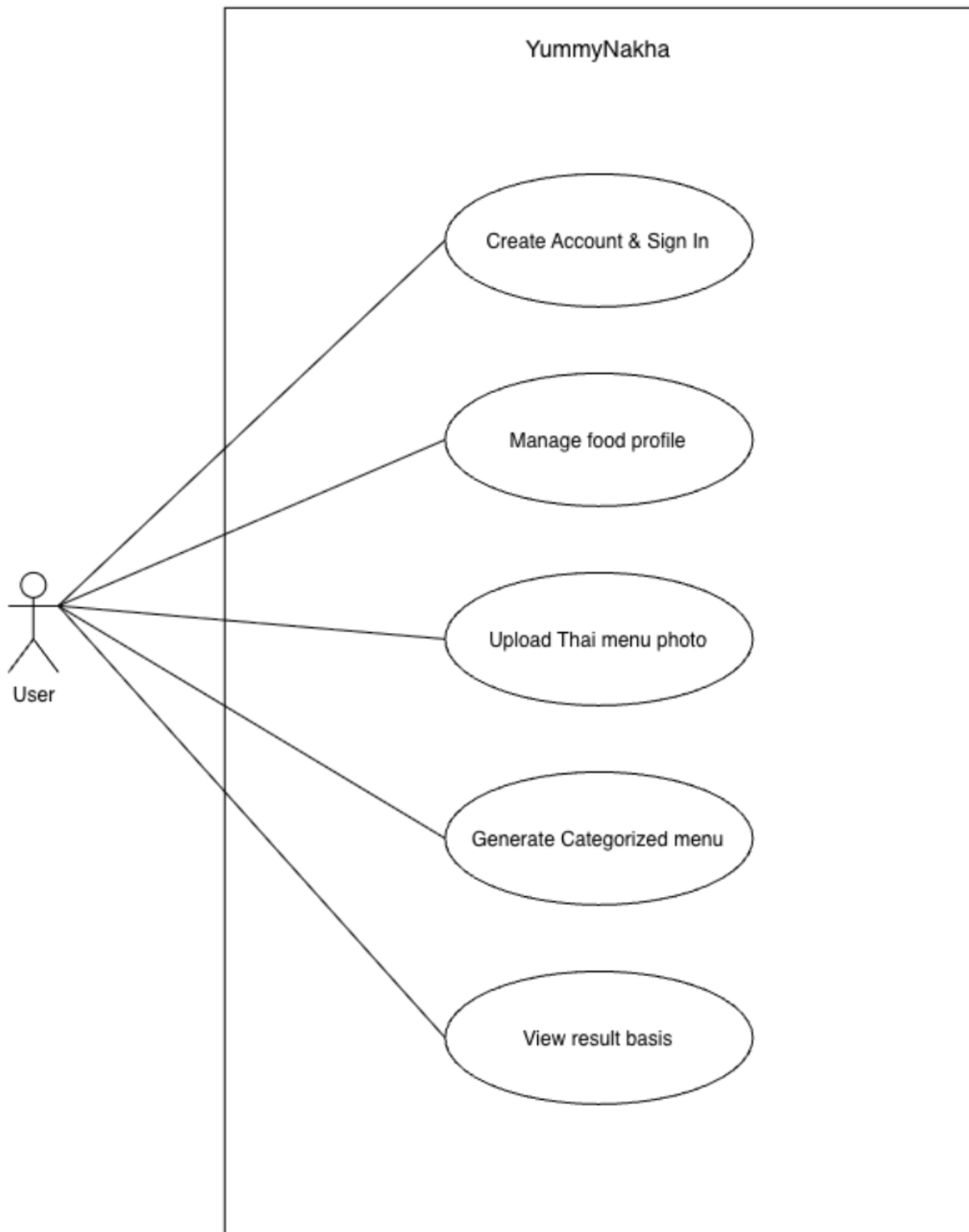
Stakeholder	Interest in the project
Target users — international students & foreigners in Thailand	Read a Thai menu, get personalized dish suggestions, and order in Thai with less guesswork.
Secondary users — diners with allergies or doctor-advised restrictions	Check a menu against their restrictions before ordering, with an honest confirm-with-staff safeguard.
Course instructors / assessors	Evidence of traceability from survey pain to requirement, feature, acceptance criterion, and test.
Third-party AI service provider (Google Gemini API)	Usage volume, cost, and rate limits; the subject of the PDPA cross-border transfer disclosure.

11. Diagrams

Context Diagram — YummyNaKha! Personalized Menu App



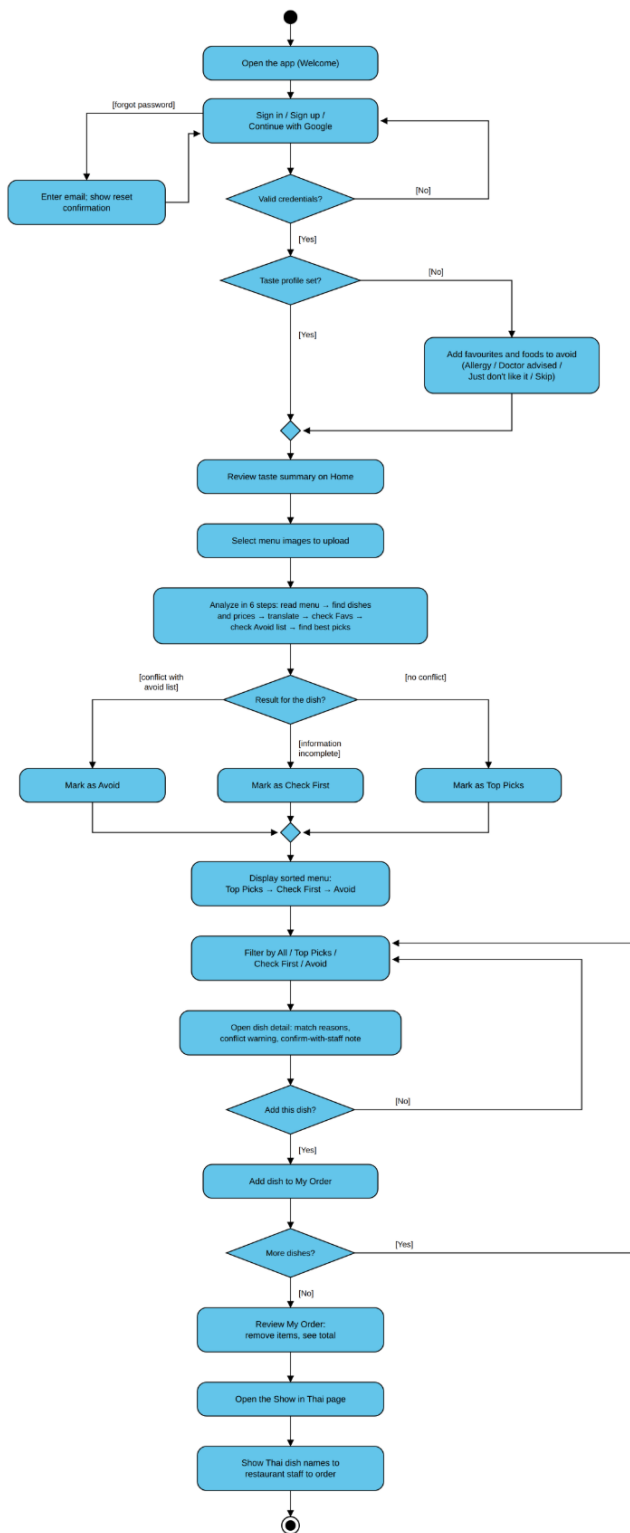
11.1. Context Diagram

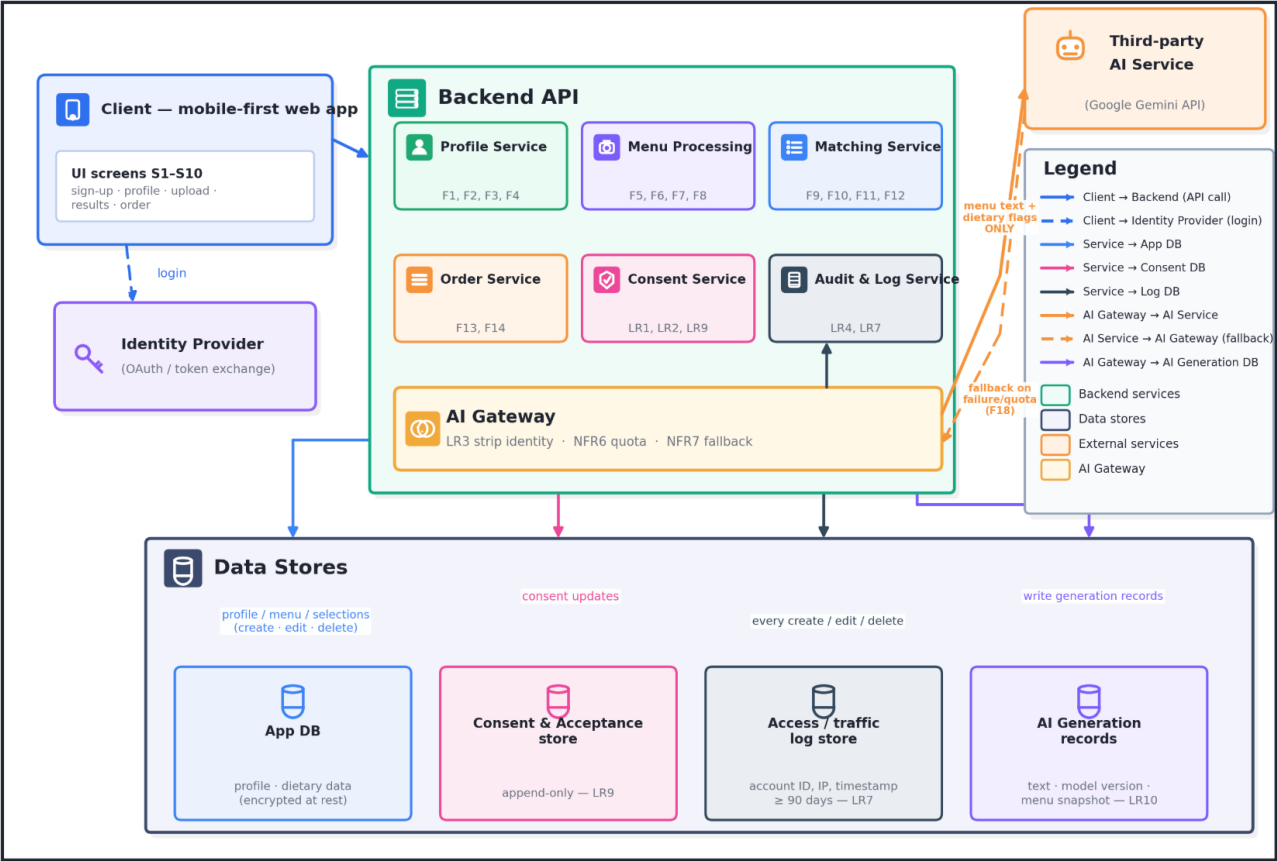


11.2. Use-Case Diagram

YummyNaKha! — Main User Activity (as built)

Figma Make prototype, v7 — menu photo to order in Thai

**11.3. Activity Diagram**



11.4. Architecture Diagram

12. Timeline and Milestones

Phase	Weeks	Milestone and deliverables
Discover	W1–W5	Problem validation; user survey/interviews; requirement spec + backlog; initial UX (4 diagrams + prototype sketch); User Validation Gate (pass/fail).
Build	W6–W8	Develop the YummyNaKha! MVP; Alpha Demo — a diner creates a food profile with allergies and restrictions, uploads a Thai menu image, receives an English menu with personalized recommendations and warnings, selects dishes, and sees them in Thai for ordering.
Test	W9–W11	Test with 15+ real users; evaluate extraction/translation accuracy and usability; Beta Review — YummyNaKha! reduces the time to find and select suitable dishes versus the diner's current method, plus feedback-driven improvements.
Deliver	W12–W14	Final improvements; working deployed system; tested final version; user-evaluation results; technical/project documentation; source code; final presentation / demo.

12.1 Deliverables checklist

- Project charter, requirement specification, and prioritised backlog (Discover).
- Context, use-case, activity, and architecture diagrams, plus the low-fidelity prototype (Discover).
- Working application covering the core workflow FE1–FE7, with FE8 and FE9 behind it (Build).
- Test evidence: the extraction/translation/matching test sets, the failure-handling checks, and the compliance log verification (Test).
- Impact report with before/after decision-time metrics, and the final portfolio pack (Deliver).

13. Risks, Constraints and Mitigations

Risk / constraint	Why it matters	Mitigation
Extraction accuracy	Menu-image quality — lighting, angle, handwriting, layout — can reduce dish/price extraction accuracy.	NFR2 targets measured on a defined test set of readable photographs; a failed read shows a plain-language message and keeps the photo on screen for crop/retry (F18, F26).
Incomplete menu information	A menu photo rarely shows every ingredient — sauces, oil, garnish, and preparation are usually invisible.	Uncertain or incomplete information always resolves to check before ordering, never recommended (F10, LR12); the confirm-with-staff instruction is the only safeguard the app can offer.
AI inference / matching errors	Ingredient inference can be wrong, and a missed allergen conflict is a safety issue, not a convenience one.	NFR4 reports missed conflicts ($\leq 5\%$) separately from false conflicts ($\leq 10\%$) on a controlled test set; every generation is stored so a failure is reproducible (LR10).
Unverifiable preparation	Actual kitchen preparation and cross-contact cannot be checked from a photo.	The product never claims a dish is safe — only that no conflicting ingredient was detected from the available information (LR12); every check/conflict tells the diner to confirm with staff.
AI cost & availability	Extraction, translation, and inference depend on a third-party service with usage costs and rate limits.	Server-side usage cap once NFR6 is set; graceful degradation to a plain-language notice, never a raw error, when the provider is down or rate-limited (NFR7).

Risk / constraint	Why it matters	Mitigation
Consent-prompt friction	A legal-sounding consent screen before the first scan could read as a wall and cause diners to abandon.	State plainly what is and is not sent, as a short list rather than a legal document (LR2, LR3).
Measurement difficulty	The project metric is decision time and steps, self-reported and hard to baseline without real users.	NFR1 defines the method up front — timed observation with 15+ real users, current method versus YummyNaKha! — rather than asserting an improvement without evidence.
Safety-wording / compliance scope creep	Consent, logging, and retention rules touch nearly every feature and can be deferred until too late.	FE9 is a cross-cutting Must feature built alongside FE1–FE8, with its own acceptance criteria, not an end-of-project audit.
Team capacity	Five members, a fixed course schedule, and a compressed Build window.	Scope locked at the end of Discover; the Build commitment is the single core workflow (FE1→FE7), and everything else waits.

14. Conclusion

YummyNaKha! addresses a common and measurable problem — the gap between a translated menu and a menu a diner can actually trust and act on — with a single, coherent workflow rather than a broad feature set. By joining translation, personalization, and an honest allergy/restriction check in one place, the system removes the dish-by-dish guesswork and the dependence on staff, friends, or translation apps that the survey evidence consistently describes.

The scope is deliberately narrow, and every requirement is traceable from a survey pain or a legal obligation through to a testable acceptance criterion. Thai PDPA, Computer Crime Act, and Electronic Transactions Act obligations are treated as build requirements from the first sprint, not as a compliance review bolted on at the end. Success will be judged on a real, measured reduction in decision time with real users — not asserted from the feature list.

15. Authorisation

The undersigned team members approve this proposal and commit to the scope, timeline, and responsibilities described above.

Student ID	Name	Role	Signature / Date
6631503089	Tay Zar Tun	Product Owner	8 Sep 2026
6631503052	Aike Paung Bra	QA / Tester	8 Sep 2026
6631503055	Chan Nyein Thu	AI Lead	8 Sep 2026
6631503093	Thiri Kyaw Khaing	Tech Lead	8 Sep 2026
6631503099	Yoon Nadi	Designer	8 Sep 2026